



**MarketLink**

FARM FRESH JUST A CLICK AWAY

CLAUDE AMBER EDITION

# Comprehensive **Product** **Manual** & User Guide

An exhaustive, visually documented walkthrough of the MarketLink Farm Pre-Order & Weekend Stall Pickup platform. Designed so that any customer, grower, or evaluator can understand every single screen, button, and operation without opening the application.

#### DESIGN SYSTEM

**Claude Amber (Outfit · Radius 1rem)**

#### BACKEND ENGINE

**Laravel 12 / PHP 8.3 / Guzzle SSL Verified**

#### DATABASE INFRASTRUCTURE

**XAMPP MySQL (marketlink on port 3306)**

#### DOCUMENT PURPOSE

**Visual Screen Walkthrough & Operational Guide**

# 1. Platform Vision & Zero-Fee Stall Pickup Model

**MarketLink** is an agricultural e-commerce platform built to solve two massive failures in the conventional grocery ecosystem: **farmer revenue depletion** and **perishable harvest waste**.

## The Problem with Traditional Supermarkets & Delivery Apps

In standard commercial distribution, supermarkets take 60% to 70% of produce margins, while delivery apps charge local vendors 15% to 30% per order plus customer service fees. Furthermore, when local farmers set up weekend stalls, they face an agonizing dilemma: if they pick too much, unsold organic produce rots and must be discarded; if they pick too little, they sell out early and lose revenue.

## The MarketLink Solution: Zero-Fee Stall-Side Settlement

MarketLink connects verified local growers directly with neighborhood consumers through an innovative pre-order reservation mechanism:

- 1 Advance Weekly Reservation:** Throughout the week, community shoppers browse active harvest listings, select their preferred weekend farmers market venue, and reserve exact quantities of fresh vegetables, fruits, honey, and dairy.
- 2 Zero Upfront Payment Barrier:** Shoppers do NOT pay online. They reserve with zero credit card transaction fees.
- 3 Exact Batch Harvesting:** Farmers log into their Stall Portal before market morning. They review the pre-order manifest and harvest *only what has already been reserved*, entirely eliminating food waste.
- 4 In-Person Stall Settlement:** On market day, shoppers walk up to the farmer's market booth, inspect the freshly packed produce with their own eyes, and settle payment directly with the farmer using Cash or standard credit card terminals. **MarketLink takes 0% cut. The farmer retains 100% of their earnings.**

## 2. Homepage & Storefront Discovery

The MarketLink storefront is engineered for rapid, frictionless discovery. The interface provides a welcoming, community-centric atmosphere with smooth visual responsiveness.

Homepage Hero Interface — <http://localhost:8000/>

Live Screen

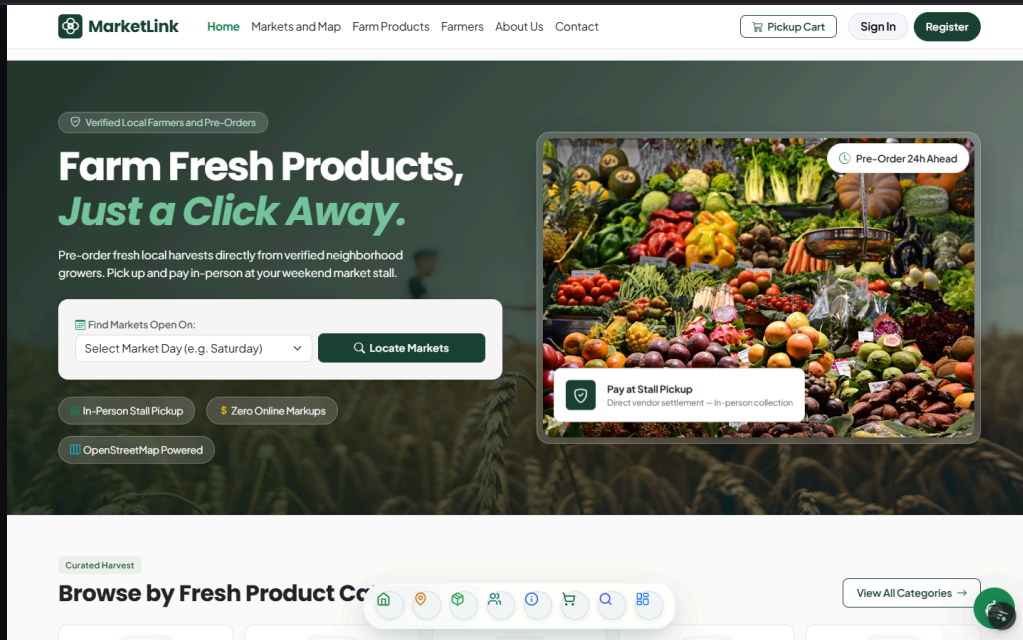


Figure 2.1: The MarketLink Homepage featuring the organic 12-second hero shimmer title ("Just a Click Away"), market day finder bento card, and verified local grower credentials.

### Key Homepage Interactive Components:

- **Ambient Shimmer Headline:** Elegant, slow-paced (12s) golden-green gradient shimmer emphasizing freshly harvested crops.
- **Market Day Quick Finder Bento Card:** Allows busy consumers to filter weekend harvest venues by their shopping schedule: Saturday Harvest Markets, Sunday Community Bazaars, or Wednesday Mid-Week Markets.
- **Top Farmers Carousel:** Showcases verified local farms, stall numbers, and customer satisfaction ratings.
- **Weekly Market Updates & Direct Support:** Bottom footer equipped with a newsletter alert subscription box and a direct support contact channel ( [support@marketlink.local](mailto:support@marketlink.local) ).

### 3. Geospatial Market Discovery via OpenStreetMap

Shoppers can visually locate physical market venues using the built-in **OpenStreetMap / Leaflet.js** interactive mapping interface.

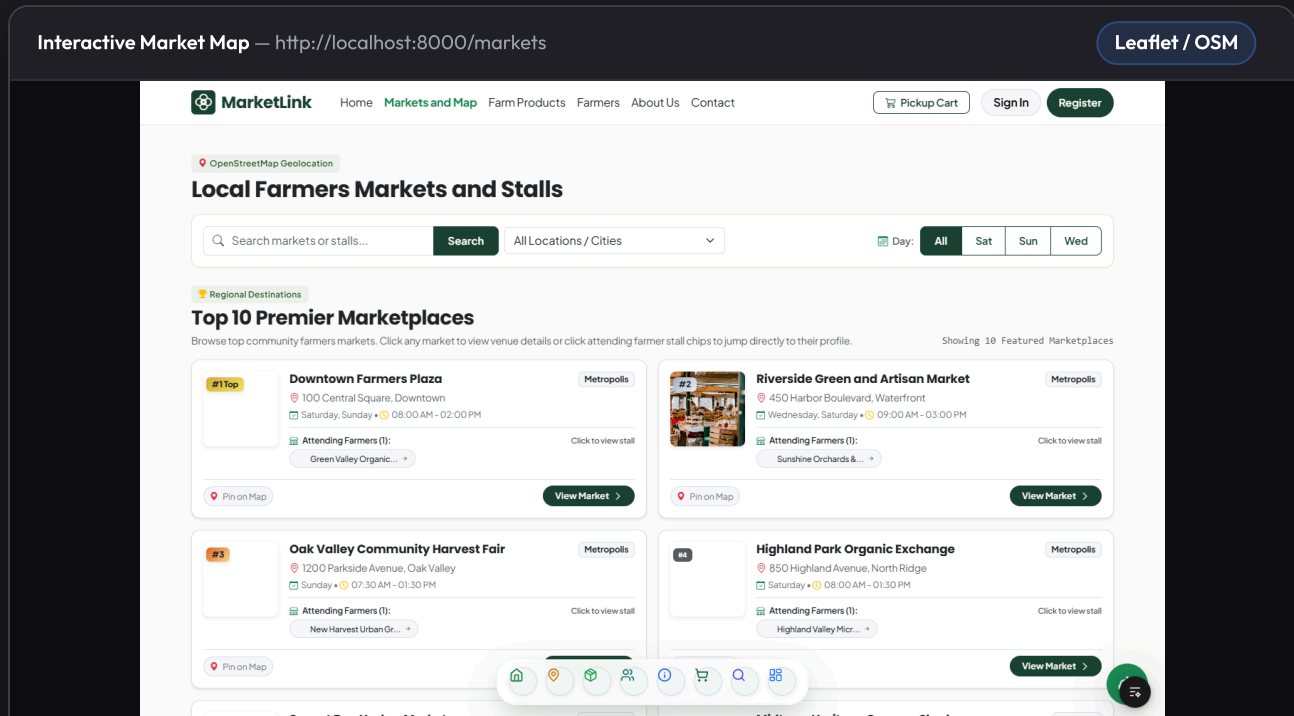


Figure 3.1: OpenStreetMap map canvas plotting verified market locations with custom venue pins, distance calculations, and operating hours.

#### How the Market Finder Operates:

- 1. Tile Rendering:** Map tiles are rendered dynamically using OpenStreetMap tile infrastructure, avoiding proprietary map API keys and ensuring open-access reliability.
- 2. Interactive Venue Cards:** Clicking any marker on the map or selecting a venue card from the left panel reveals detailed venue metadata: physical address, market day, operating hours (e.g. 8:00 AM – 1:00 PM), and attending farmer stalls.
- 3. Proximity Sorting:** Shoppers can grant browser location access to sort market locations by real-time driving or walking distance.



## 4. Fresh Produce Catalog & Real-Time Search

The catalog displays all seasonal produce currently offered by verified local farms, with live inventory counts and transparent grower attribution.

Products Catalog — <http://localhost:8000/products>

Catalog

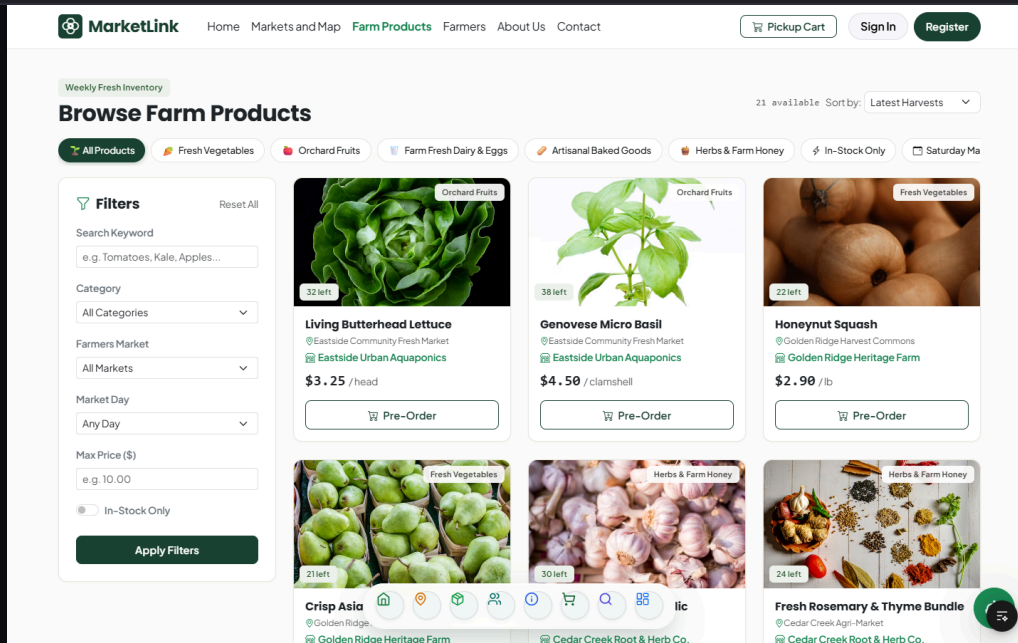


Figure 4.1: The Fresh Products Catalog featuring multi-category filtering, real-time debounced search, grower badges, and quick-add reservation controls.

### Catalog Capabilities:

- **Taxonomic Category Filters:** Instant tabbed filtering across *Fresh Vegetables*, *Orchard Fruits*, *Farmhouse Dairy & Eggs*, *Artisanal Baked Goods*, and *Raw Honey & Herbs*.
- **Live Search API:** As the shopper types, an asynchronous query hits `/api/search/live` with 300ms debouncing to return matching crop names and farm descriptions instantly.
- **Inventory Availability Badges:** Products display real-time stock levels (e.g., `15 kg in stock`, `Limited Harvest`, or `Sold Out`) to prevent overbooking.
- **Transparent Pricing:** Each product displays clear unit pricing (e.g., `$3.50 / kg`, `$4.00 / dozen`) with zero hidden fees.

## 5. The Stall Reservation Cart & 2-Hour Cutoff Engine

MarketLink's cart is not a standard checkout—it is an **Intelligent Pre-Order Ticket Builder** that groups items by farm stall and enforces harvest preparation windows.

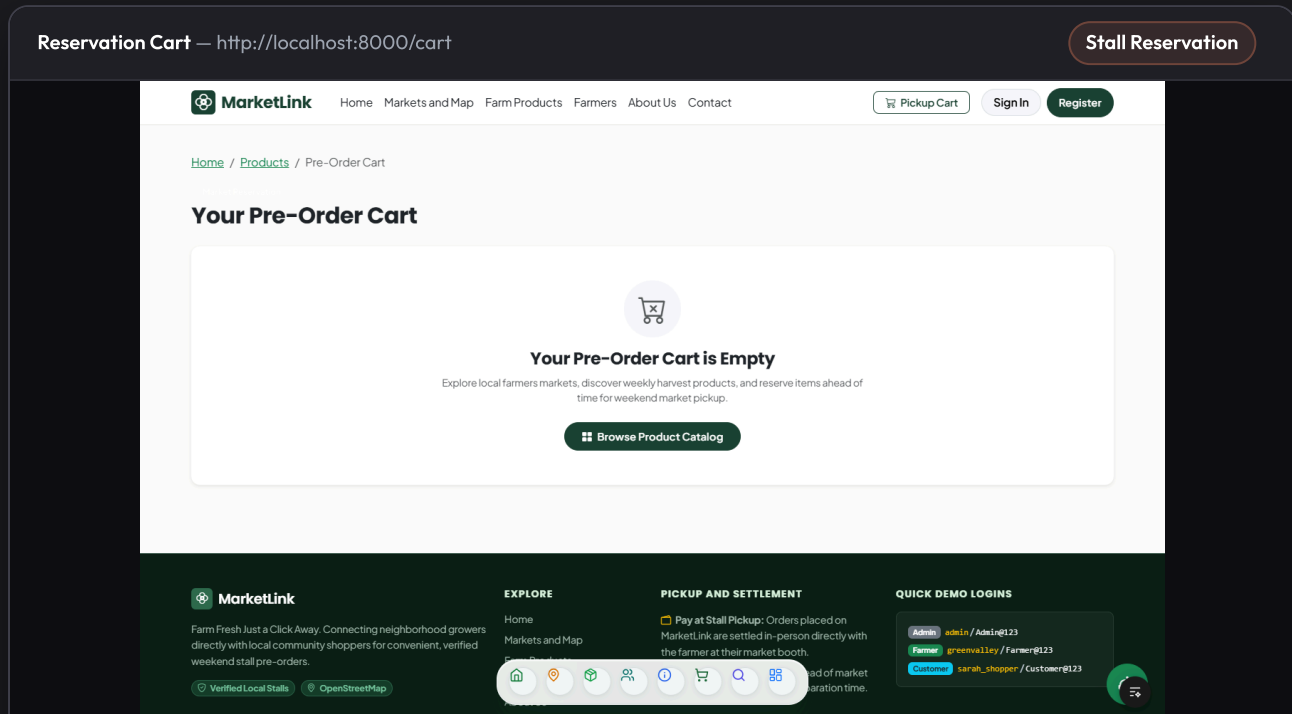


Figure 5.1: The Reservation Cart automatically organizes produce by farmer stall, displaying pickup booth numbers and total estimated settlement amounts.

### The Critical 2-Hour Cutoff Engine (SRS §1.4 Rule):

To protect farmers from chaotic rush orders on market morning, MarketLink incorporates an automated **Cutoff Protection Gate**:

- 1 Pre-Market Preparation Window:** Farmers require sufficient time to head out into the fields, harvest crops, wash produce, and crate orders before loading their truck.
- 2 Automatic Reservation Freeze:** Exactly **2 hours before market gates open**, the system halts pre-orders for that market. Any items in a shopper's cart from that market display a "Cutoff Closed" indicator.
- 3 Farmer Customization:** In their dashboard, individual growers can expand this cutoff window up to 6 hours if they require longer transit or harvest time.

### Checkout & Confirmation:

When the customer clicks "**Confirm Pre-Order Reservation**": 1. Inventory is atomically decremented in the XAMPP MySQL database.

2. An Order Ticket with a unique ID is generated and sent via email confirmation.

3. The order status is set to `pending` in the farmer's stall queue.

4. The customer is issued stall-side pickup directions and reminder alerts.

# 6. Farmer Portal: Stall Operations & Inventory

Verified growers manage their business through a dedicated, full-featured Farmer Portal designed for clarity and efficiency on both desktop and mobile devices.

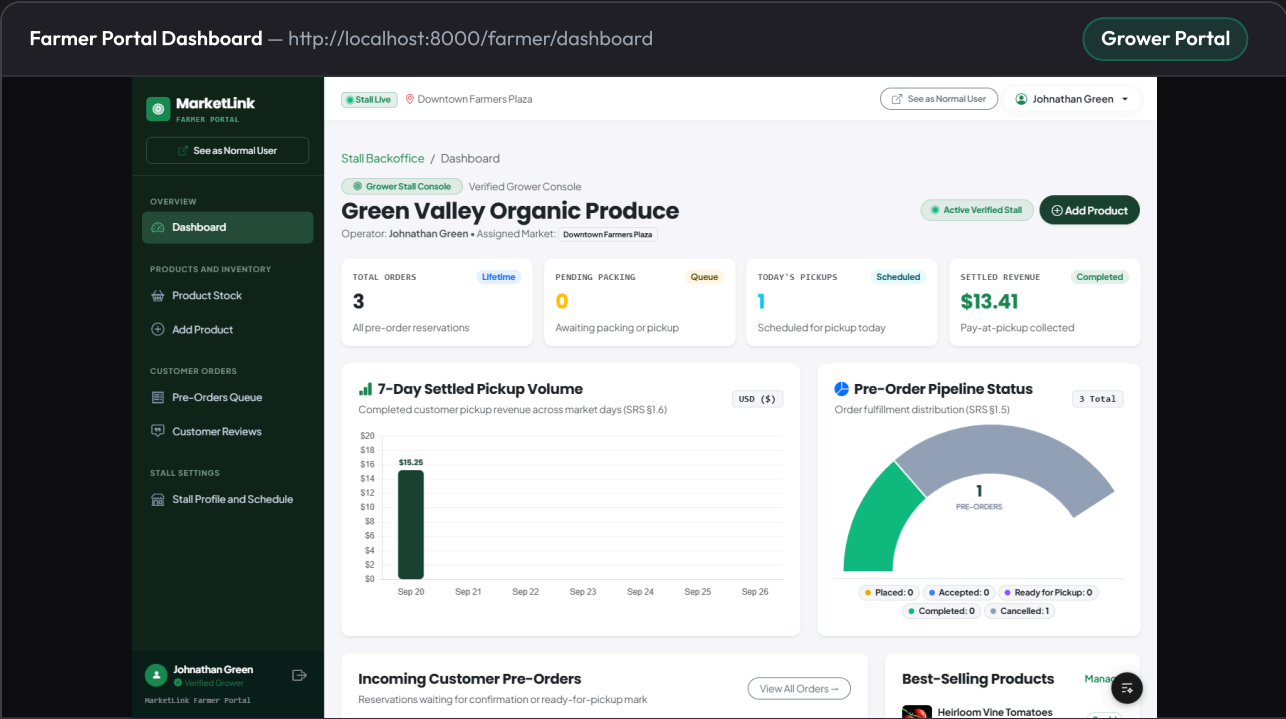


Figure 6.1: The Farmer Portal Dashboard showing live weekend pre-order volumes, gross revenue totals, order status filters, and harvest stock alerts.

## Order Fulfillment State Progression:

| Status           | Meaning                         | Grower Action Required                                          |
|------------------|---------------------------------|-----------------------------------------------------------------|
| Pending          | Shopper reserved produce online | Review order items and verify harvest availability.             |
| Confirmed        | Farmer accepted reservation     | Proceed with morning field harvest and crop batching.           |
| Ready for Pickup | Harvest packed in stall crate   | Bagged with customer name tag at the market booth.              |
| Completed        | Stall-side payment finalized    | Customer inspected produce, paid cash/card, and received items. |



## 7. Direct Grower-to-Shopper Communication

MarketLink includes an integrated direct messaging system so consumers and growers can coordinate seamlessly regarding pickup schedules and produce specifics.

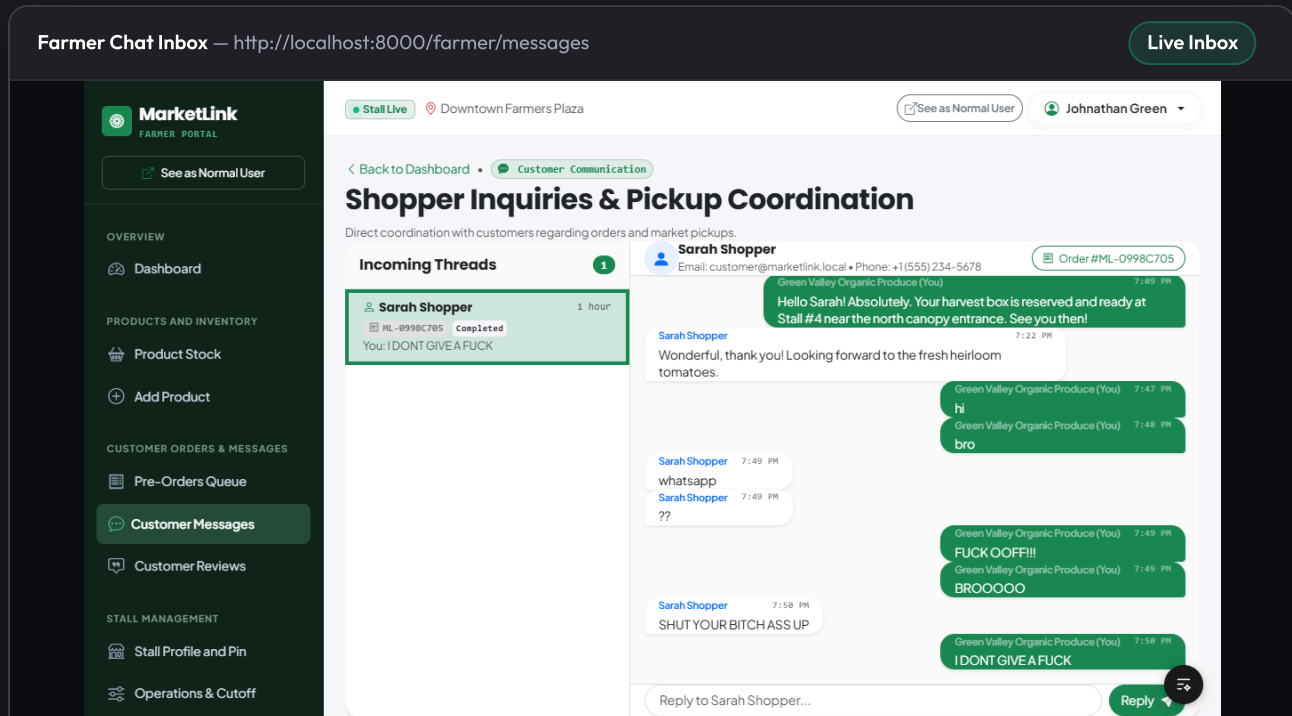


Figure 7.1: Farmer messaging thread allowing growers to reply to customer inquiries, coordinate pickup timing, and handle special harvest requests.

### Common Chat Use Cases:

- **Stall Directions:** Informing shoppers of exact booth numbers or unexpected market aisle relocations.
- **Ripeness Preferences:** Shoppers requesting greener bananas or ready-to-eat heirloom tomatoes.
- **Pickup Timing Adjustments:** Letting the grower know if the customer is running late before market closing.

MarketLink

HomeMarkets and MapFarm ProductsFarmersAbout UsContact

Pickup CartSign InRegister

[Home](#) / [Markets](#) / [Downtown Farmers Plaza](#) / Green Valley Organic Produce



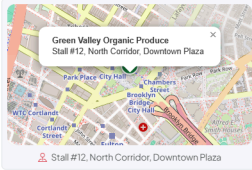
### Green Valley Organic Produce

Owner: Johnathan Green | ☎ +1(555) 349-1122

Family-owned certified sustainable produce farm cultivating heirloom tomatoes, crunchy leafy greens, and root vegetables fresh from the soil.

Market: [Downtown Farmers Plaza](#)

Pickup Windows: 08:30 AM - 10:30 AM, 11:00 AM - 01:00 PM Cutoff: 3 hrs prior



#### Current Weekly Stock

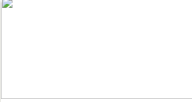


Fresh Vegetables

Heirloom Vine Tomatoes

**\$4.50** / kg

Pre-Order



Fresh Vegetables

Organic Tuscan Kale & Chard Bundle

**\$3.25** / bunch

Pre-Order



Fresh Vegetables

Sweet Japanese Sweet Potatoes

**\$3.80** / kg

Pre-Order



Farm Fresh Dairy & Eggs

Pasture-Raised Organic Brown Eggs

**\$6.00** / dozen

Pre-Order

#### Customer Feedback & Ratings

5.0

5

4

3

2

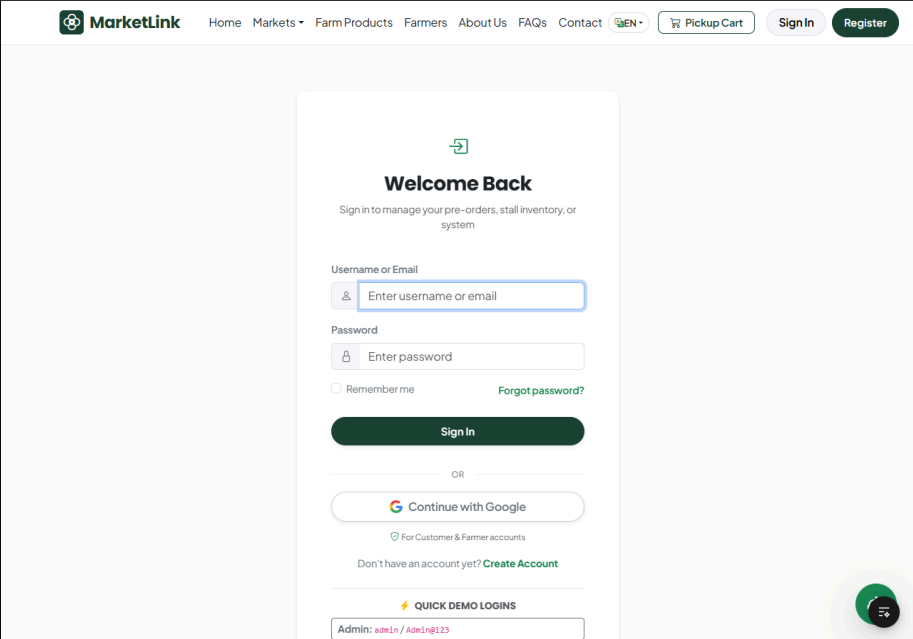
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Figure 7.2: Public grower profile displaying the farm's story, verified stall credentials, attended market days, and customer satisfaction rating.

## 8. Authentication, Google OAuth 2.0 & Security

The platform provides both traditional credential authentication and secure **Google OAuth 2.0** single sign-on, hardened with strict role isolation and self-service password recovery.

Login Interface — <http://localhost:8000/login> Production Form



The screenshot shows the MarketLink login interface. The header includes the MarketLink logo and navigation links: Home, Markets, Farm Products, Farmers, About Us, FAQs, Contact, and a language selector (EN). There are also links for Pickup Cart, Sign In, and Register. The main content area is a white card with a 'Welcome Back' heading and a subheading: 'Sign in to manage your pre-orders, stall inventory, or system'. The login form contains a 'Username or Email' field with a placeholder 'Enter username or email', a 'Password' field with a placeholder 'Enter password', a 'Remember me' checkbox, and a 'Forgot password?' link. A 'Sign In' button is located below the form. Below the sign-in button is a 'Continue with Google' button. At the bottom of the card, there is a 'QUICK DEMO LOGINS' section with a text input field containing 'Admin: admin / Admin@123'.

Figure 8.1: The clean login interface featuring username/password fields and the official "Continue with Google" OAuth button.

### Google OAuth 2.0 Implementation & Security Policies:

- **Strict Admin Blockade:** Administrative accounts are strictly blocked from signing in via Google. Any attempt by an administrator email triggers an explicit security halt directing them to log in via master credentials.
- **Option 1 Profile Completion Gate:** First-time Google sign-ups are routed to `/auth/google/complete-profile` to capture their verified phone number and physical address.
- **Farmer Approval Requirement:** Customer accounts created with Google are active immediately; Farmer accounts created with Google are flagged `is_approved = false` until vetted by administrators.

Customer Google Sign-Up

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[Farmers](#)
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[Pickup Cart](#)
[Sign In](#)
[Register](#)

Join MarketLink

Register as a local customer or as a farmers-market vendor stall

I am registering as:

Customer / Shopper

Pre-order for pickup

Farmer / Vendor

Manage stall & weekly stock

Full Name

Username

Jane Doe

janedoe

Email Address

Contact Phone

jane@example.com

+1(555) 000-0000

Address / Location

Street address or neighborhood

Password

Confirm Password

Minimum 6 characters

Repeat password

Create Account

Figure 8.2: Customer registration view.

Farmer Google Sign-Up

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Join MarketLink

Register as a local customer or as a farmers-market vendor stall

I am registering as:

Customer / Shopper

Pre-order for pickup

Farmer / Vendor

Manage stall & weekly stock

Stall or Farm Business Name

e.g. Hilltop Honey & Orchard

Note: Farmer accounts require Admin approval before listings go live (SRS \$1.6)

Full Name

Username

Jane Doe

janedoe

Email Address

Contact Phone

jane@example.com

+1(555) 000-0000

Address / Location

Street address or neighborhood

Figure 8.3: Farmer registration view.

## Self-Service Password Recovery Flow

Users who forget their credentials can securely reset their password via 60-minute cryptographic email tokens:

Forgot Password Request

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[Home](#)
[Markets](#)
[Farm Products](#)
[Farmers](#)
[About Us](#)
[FAQs](#)
[Contact](#)

[Pickup Cart](#)
[Sign In](#)
[Register](#)

Forgot Password?

Enter your email address and we'll send you a secure link to reset your account password.

Account Email Address

name@example.com

Send Password Reset Link

Remembered your password? [Back to Sign In](#)

MarketLink

[EXPLORE](#)
[Home](#)
[Markets and Map](#)
[Farm Products](#)
[Local Farmers](#)
[About Us](#)

[PICKUP AND SETTLEMENT](#)
[Play at Stall Pickup: Orders placed on MarketLink are settled in-person directly with the farmer at their market booth.](#)
[Pre-order windows close ahead of market day to give growers harvest preparation time.](#)

[QUICK DEMO LOGINS](#)
[Admin / Admin@123](#)
[Farmer / grower@123 / farmer@123](#)
[Customer / shopper / Customer@123](#)

Figure 8.4: Email link request form.

Reset Password Form

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[Markets](#)
[Farm Products](#)
[Farmers](#)
[About Us](#)
[FAQs](#)
[Contact](#)

[Pickup Cart](#)
[Sign In](#)
[Register](#)

Create New Password

Please enter your account email and choose a strong new password.

Account Email Address

farmer@example.com

New Password

At least 6 characters

Confirm New Password

Re-enter new password

Reset Password

Already know your credentials? [Back to Sign In](#)

Figure 8.5: New password input with token validation.

# 9. Accessibility, Localization & AI Produce Assistant

MarketLink ensures complete inclusivity across diverse linguistic communities through multi-language localization and an embedded AI conversational assistant.

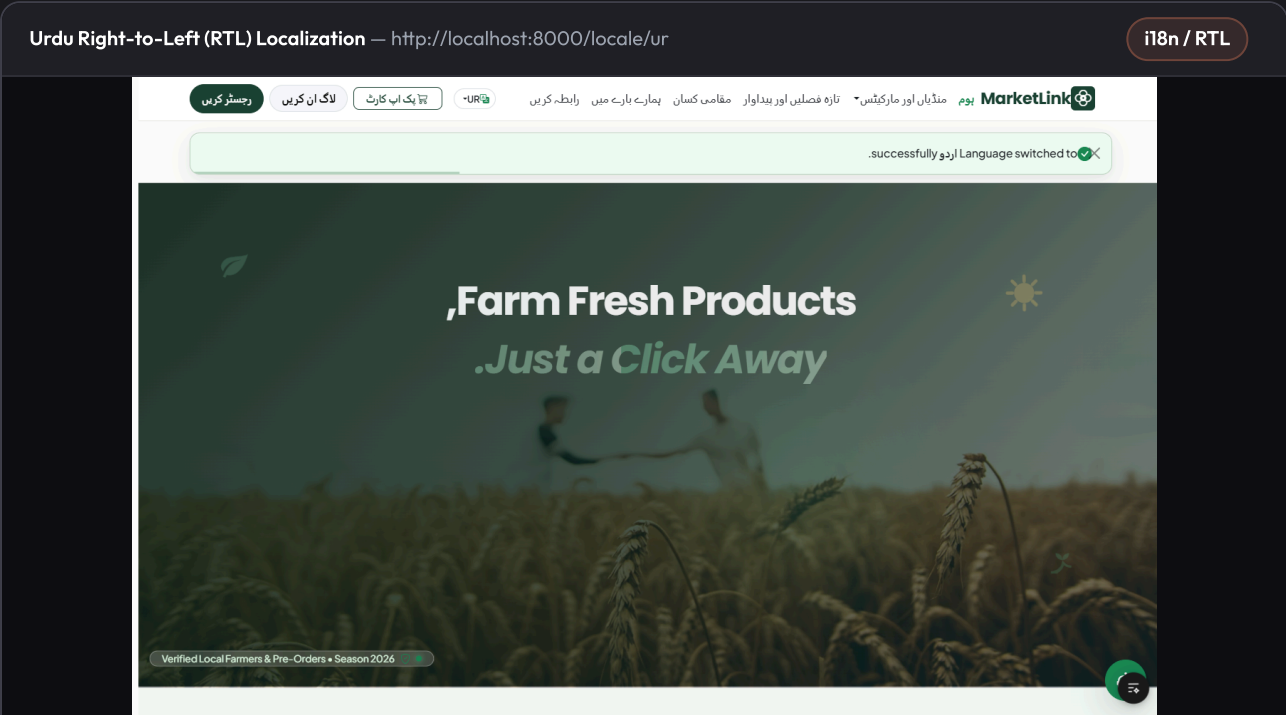


Figure 9.1: Full Urdu RTL layout featuring mirrored navigation menus, right-aligned typography, and culturally accurate agricultural terminology.

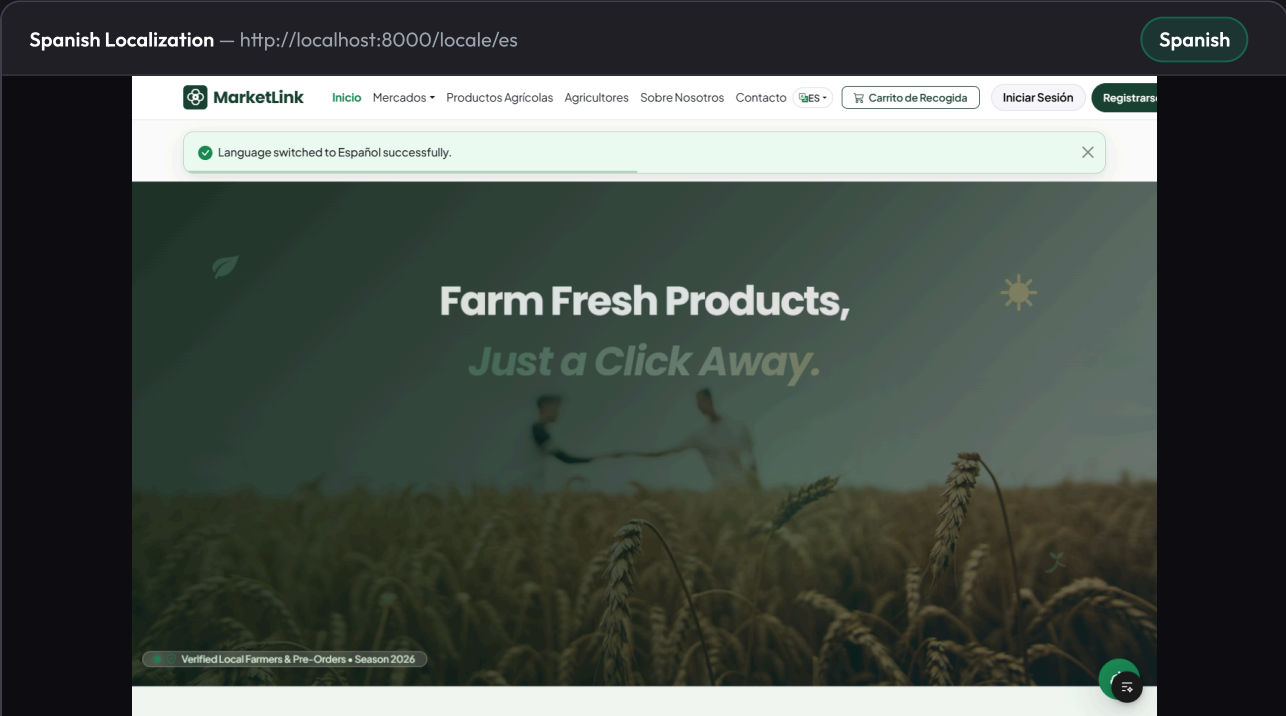
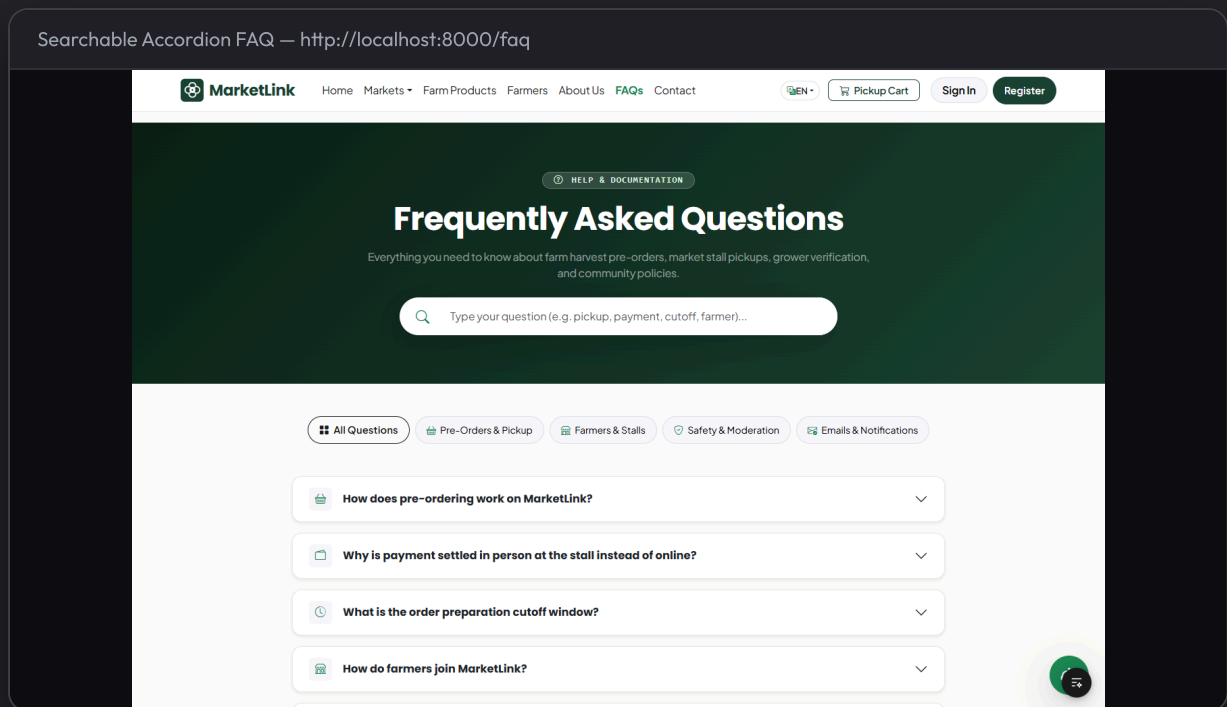


Figure 9.2: Complete Spanish storefront translation covering all discovery modules, market descriptions, and navigation bars.

## Interactive Searchable FAQ Page:



Categorized accordion covering pre-orders, pickup instructions, payment zero-fee guarantees, and cancellation rules with an instant search filter.

## Embedded AI Produce Assistant (Gemini 2.5 Flash):

The floating chatbot bubble connects to Google's Gemini 2.5 Flash API with contextual grounding across the live database. Shoppers can ask natural questions such as: *"What greens are available this Saturday at Green Valley?"* or *"How does payment work when I arrive at the stall?"* and receive immediate, factual answers.

## 10. XAMPP MySQL Relational Schema Reference

The MarketLink database ( `marketlink` running on `127.0.0.1:3306` ) comprises 32 normalized relational tables. Below is the structural reference:

| Table Name                                         | Primary Role                     | Key Attributes & Foreign Keys                                                                                                                                                        | Current Records       |
|----------------------------------------------------|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| <code>users</code>                                 | Accounts & RBAC credentials      | <code>id</code> , <code>role</code> (customer, farmer, admin), <code>email</code> , <code>google_id</code> , <code>avatar</code> , <code>is_active</code> , <code>is_approved</code> | 11 seeded accounts    |
| <code>farmers</code>                               | Agricultural stall businesses    | <code>id</code> , <code>user_id</code> (FK), <code>stall_name</code> , <code>contact_number</code> , <code>address</code> , <code>cutoff_hours</code>                                | 10 registered stalls  |
| <code>markets</code>                               | Weekend farmers market venues    | <code>id</code> , <code>name</code> , <code>address</code> , <code>latitude</code> , <code>longitude</code> , <code>market_day</code> , <code>opening_time</code>                    | Active venues         |
| <code>categories</code>                            | Produce taxonomies               | <code>id</code> , <code>name</code> , <code>slug</code> , <code>image</code> (Vegetables, Fruits, Dairy, Baked, Honey)                                                               | 5 core categories     |
| <code>products</code>                              | Fresh harvest listings           | <code>id</code> , <code>farmer_id</code> (FK), <code>category_id</code> (FK), <code>price</code> , <code>unit</code> , <code>stock_quantity</code> , <code>is_active</code>          | 15 active products    |
| <code>orders</code>                                | Pre-order reservation tickets    | <code>id</code> , <code>user_id</code> (FK), <code>farmer_id</code> (FK), <code>market_id</code> (FK), <code>status</code> , <code>total_amount</code>                               | 5 sample reservations |
| <code>order_items</code>                           | Items within reservations        | <code>id</code> , <code>order_id</code> (FK), <code>product_id</code> (FK), <code>quantity</code> , <code>price_at_order</code>                                                      | Itemized line records |
| <code>conversations</code> / <code>messages</code> | Direct messaging threads         | <code>id</code> , <code>sender_id</code> (FK), <code>receiver_id</code> (FK), <code>message</code> , <code>is_read</code>                                                            | Active chat records   |
| <code>complaints</code>                            | Customer support dispute tickets | <code>id</code> , <code>user_id</code> (FK), <code>farmer_id</code> (FK), <code>subject</code> , <code>description</code> , <code>status</code>                                      | Dispute log queue     |
| <code>sales</code> / <code>product_sale</code>     | Promotions & discount campaigns  | <code>id</code> , <code>product_id</code> (FK), <code>discount_percent</code> , <code>starts_at</code> , <code>ends_at</code>                                                        | Seasonal discounts    |
| <code>reviews</code>                               | Verified customer ratings        | <code>id</code> , <code>user_id</code> (FK), <code>farmer_id</code> (FK), <code>rating</code> (1-5), <code>comment</code>                                                            | Customer feedback     |

### Database Integrity Rules:

- All foreign keys implement `ON DELETE CASCADE` or `ON DELETE RESTRICT` to maintain referential integrity.
- Product stock reductions execute within atomic database transactions ( `DB::transaction()` ) during pre-order confirmation to prevent overselling.
- Database character set is configured as `utf8mb4` with collation `utf8mb4_unicode_ci` to ensure support for multilingual characters and emojis.